

5 Ah to 100 Ah.

Dusk-to-dawn sensor. Normally, in a solar-photovoltaic-based installation—for example, solar home lighting system, solar lantern or solar street-light—the load (the light) is switched on at dusk (evening) and switched off at dawn (morning). During daytime, the load is disconnected from the battery and the battery is recharged with current from the solar panel. The micro-controller needs to know the presence of the solar panel voltage to decide whether the load is to be connected to or disconnected from the battery, or whether the battery should be in charging mode or discharging mode. A simple sensor circuit is built using a potential divider formed around resistors R8 and R9, zener diode ZD1 and transistor T1 for the presence of panel voltage.

Charge control. Relay RL1 connects the solar panel to the battery through diode D1. Under normal conditions, it allows the charging current from the panel to flow into the battery. When the battery is at full charge (14.0V), the charging current becomes 'pulsed'. To keep the overall current consumption of the solar controller low, normally-closed (N/C) contacts of the relay are used and the relay is normally in de-energised state.

Load control. One terminal of the load is connected to the battery through fuse F1 and another terminal of the load to an n-channel power MOSFET T3. MOSFETs are voltage-driven devices that require virtually no drive current. The load current should be limited to 10A. One additional MOSFET is connected in parallel for more than 10A load current.

Circuit description

Basically, there are two methods of controlling the charging current: series regulation and parallel (shunt) regulation. A series regulator is inserted between the solar panel and the battery. The series type of regulation 'wastes' a lot of energy while charging the battery as the control circuitry is always active and series regulator requires

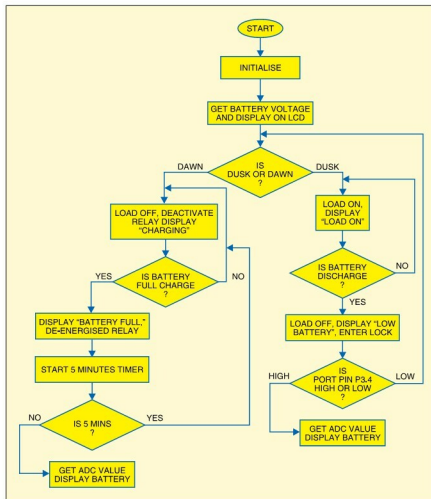


Fig. 5: Flow-chart of the source program

the input voltage to be 3-4 volts higher than the output voltage. The current and voltage output of a solar panel is governed by the angle of incidence of light, which keeps varying.

Parallel regulation is preferred in solar field. In parallel regulation, the control circuitry allows the charging current (even in mA) to flow into the battery and stop charging once the battery is fully charged. At this stage, the charging current is wasted by converting into heat (current is passed through low-value, high-wattage resistor); this part of the regulation dissipates a lot of heat.

In this project, we have used parallel regulation technique but instead of wasting the charging current as heat, we have made it pulsed and applied to the battery to keep the battery

topped-up.

After power-on, the microcontroller reads the battery voltage with the help of the ADC and displays the values on the LCD. It monitors the input signal from the dusk-to-dawn sensor and activates the load or charging relay RL1 accordingly. The digital voltmeter works up to 20V. As V_{ref} of the ADC is connected to V_{CC} (5V), the input voltage to the ADC cannot exceed +5V. A potential divider is used at pin 2 of the ADC (IC2) using resistors R5, R6 and R7 to scale down the voltage from 0V-20V to 0V-0.5V. The ADC output is multiplied four times and displayed on the LCD as battery voltage.

When the solar panel voltage is present, the dusk-to-dawn sensor provides a signal to the microcontroller, which then displays 'Charging' mes-